

MUSA RIOBA

Data Engineer · Author · Technical Trainer

Nairobi, Kenya | nyakerabachi@gmail.com | github.com/musanyaks | linkedin.com/in/musarioba

PROFESSIONAL SUMMARY

Data Engineer with a BSc in Actuarial Science and 6+ years of hands-on experience across freelancing, technical training, and open-source development. Author of "Production Data Science with R," an 8-chapter guide to production-grade ML pipelines, ETL orchestration, real-time fraud detection, interactive dashboards, and A/B testing frameworks. Builds systems that make data reliable, scalable, and actionable — from CDISC-standard clinical trial dashboards to national census data packages used across three countries.

SELECTED PROJECTS

End-to-End Data Engineering Pipeline — *Python, Apache Airflow, Apache Spark, dbt, Snowflake, Docker*

- Architected 3 Airflow DAGs using CeleryExecutor with task dependency graphs, XCom, and automatic retry policies
- Built dual-mode processing — Pandas for <1GB datasets, PySpark for 1–50GB with Adaptive Query Execution
- Implemented dbt medallion architecture — staging → intermediate → incremental marts with SCD Type 2 snapshots
- Developed multi-source extractors — CSV, Snowflake (key-pair auth), REST API with retry logic
- Built data quality gates — runtime validators, dbt schema tests, structured JSON logging via structlog
- Containerized full stack with Docker Compose — Airflow + Spark + PostgreSQL + Redis
- Set up CI/CD — GitHub Actions, pre-commit hooks, pytest with unit/integration/Spark test matrix

TrialView Pro — Clinical Trial Monitoring Dashboard — *R Shiny, SQLite, Docker, CI/CD*

- Built a production-ready Phase II diabetes clinical trial monitoring system on CDISC SDTM-standard data, with role-based access control for Investigator, Monitor, and Admin roles
- Implemented enrollment and safety monitoring (AE/SAE tables, SOC drill-down, CSV export), efficacy tracking, and longitudinal patient management
- Added a full SQLite-backed audit trail for compliance and deployed via Docker and Shinyapps.io with a GitHub Actions CI/CD pipeline (linting, styling, data-integrity tests)

Sales Analytics Dashboard — Business Intelligence Tool — *R Shiny, bslib, Plotly, DT*

- Delivered a production BI dashboard for sales analytics and executive reporting with reactive filters (region, product, time period) driving live KPI cards and Plotly charts
- Built a sortable transactions table with styled status badges (DT) and one-click CSV export, in a fully responsive dark-theme UI

Kenya Insurance Hub — *R Shiny, bslib, Plotly, DT*

- Built insurance marketplace platform with 4 premium calculators (Motor, Health, Life, Agriculture) across 28 insurers
- Created company directory with 28 licensed insurers, ratings, contacts, branches, and 69 products

- Developed market analysis with interactive bar, pie, and treemap charts for market share and product distribution
- Integrated NHIF benefits coverage for 19 benefits with co-payment and waiting period details

Census Data Packages — `kenyacensus`, `ugcensus`, `sacensus` — R Packages, Open Source

- Published three open-source R packages providing toolkits for accessing, cleaning, and visualizing national census data for Kenya (8 census years, 47 counties, 13 datasets), Uganda (4 census years, 17 sub-regions), and South Africa (4 census years, 9 provinces)
- Built in multi-level geography support, population pyramids, growth-rate projections, and time-series comparison tooling; `kenyacensus` released under a CC0 public-domain license

PUBLISHED WORK

"Production Data Science with R" — Author

An 8-chapter guide to building production-grade data systems in R, covering end-to-end ML pipelines, automated ETL with targets, real-time fraud detection with Redis and isolation forests, demand forecasting (Prophet + XGBoost ensembles), customer segmentation (RFM, K-Means), marketing attribution (Markov chains), interactive Shiny dashboards, and sequential/Bayesian A/B testing frameworks — organized around reproducibility, scalability, observability, and safety.

PROFESSIONAL EXPERIENCE

Freelance Data Engineer & Analyst | Upwork & Independent | 2020 – Present

- Delivered custom data analysis and visualization projects for clients across healthcare, demographics, and research sectors
- Built automated data pipelines using R and Python to extract, clean, and structure datasets from APIs, web sources, and raw files
- Developed interactive dashboards for client reporting, enabling stakeholders to explore data without technical expertise
- Created reusable data tools and scripts that reduced manual processing time for recurring client workflows
- Managed end-to-end project delivery: requirements gathering, data processing, visualization, and client presentation

Technical Trainer — Data Analysis & Programming | 2020 – 2024

- Taught data analysis, R, Python, and production best practices to 100+ students and professionals
- Developed curricula and training materials covering data cleaning, statistical analysis, dashboard building, and reproducible research
- Mentored junior analysts on best practices in data structuring, pipeline design, and code documentation
- Conducted hands-on workshops on building Shiny dashboards, creating R packages, and automating data workflows

TECHNICAL SKILLS

Languages: R (Expert), Python, SQL, Bash, HTML/CSS

Data Engineering & Pipelines: Apache Airflow, Apache Spark, dbt, targets (ETL orchestration), ETL/ELT pipelines, data cleaning, data modeling, web scraping, API integration

Machine Learning & Analytics: tidymodels, XGBoost, Prophet, K-Means / RFM, Markov chains, isolation forests

Dashboards & Visualization: R Shiny, ggplot2, Plotly, shinydashboard, DT (DataTables), Matplotlib

Databases & Cloud: Snowflake, PostgreSQL, Redis, BigQuery, SQLite, dbplyr, AWS (learning)

Production & Deployment: Docker Compose, GitHub Actions, Shinyapps.io, plumber (APIs), Git/GitHub, pre-commit, bookdown

Experimentation & Testing: A/B testing, sequential testing, Bayesian methods, statistical modeling

Package Development: R package dev (devtools, roxygen2), Documentation, MIT Licensing, CC0

EDUCATION

BSc, Actuarial Science | Karatina University

Strong foundation in statistical modeling, probability theory, risk analysis, and financial mathematics.

Coursework in data analysis, econometrics, and computational methods.